

Physics of Complex Systems Homework 9

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Problem 1 (Entropy of the Binomial Distribution). The binomial distribution reads

$$P_n(k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

- (a) Use Mathematica[®] or similar software packages to compute the Shannon entropy (measured in bit) for $p = \frac{1}{2}$ and $n = 1, 2, 4, 8$.
- (b) Approximate the factorials by Stirling's formula, and Taylor-expand the resulting approximation in its maximum. Use this result to approximate the binomial distribution for $n \rightarrow \infty$ and $p = \frac{1}{2}$ by a normal distribution.
- (c) Compute the entropy of the approximated distribution, replacing the sum by a continuous integration. How does it grow with n ?

Proof. (a) The Shannon entropy is given by

$$H(X) = \mathbb{E}[-\log f(X)] = - \int_X f(x) \log f(x) dx.$$

where X is a discrete random variable, and hence the integral is an integral over the counting measure, i.e. a sum.

$$H(x) = - \sum_{k=0}^n \binom{n}{k} p^k (1-p)^{n-k} \log \left[\binom{n}{k} p^k (1-p)^{n-k} \right].$$

We evaluate this with the following python code:

```
1 import numpy as np
2 from math import comb
3
4
```

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5 def binomial_entropy(n, p):
6     ans = 0
7     for k in range(n+1):
8         Pk = comb(n, k) * (p**k) * ((1-p)**(n-k))
9         ans -= Pk * np.log2(Pk)
10    return ans

```

which yields the results

$$S(1) = 1$$

$$S(2) = \frac{3}{2}$$

$$S(4) = 2.0306390622295662$$

$$S(8) = 2.544197507453808$$

(b) The binomial coefficient is given by

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Since we are sending n to infinity, we have to take k and $n - k$ large. This works, since the rest of the distribution will tend to a set of measure zero. Thus, we apply Stirling's formula

$$n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n.$$

Substituting this in, we find that

$$\begin{aligned} \binom{n}{k} &= \frac{1}{\sqrt{2\pi}} \sqrt{\frac{n}{k(n-k)}} \frac{(n/e)^n}{(k/e)^k ((n-k)/e)^{n-k}} \\ &= \frac{1}{\sqrt{2\pi}} \sqrt{\frac{n}{k(n-k)}} \frac{n^n}{k^k (n-k)^{n-k}} \end{aligned}$$

Substituting, we find that

$$P_n(k) = \frac{1}{\sqrt{2\pi}} \sqrt{\frac{n}{k(n-k)}} \frac{n^n}{k^k (n-k)^{n-k}} p^k (1-p)^{n-k}.$$

In expectation that the distribution will be peaked around $k = np$, we write $k = np + x$. Note that we assume that np is integer. Since we are taking a limit, we can

consider the subsequence of n for which $np \in \mathbb{Z}$, which will yield the same limit. Then, the distribution simplifies to

$$P_n(k) = \frac{1}{\sqrt{2\pi}} \sqrt{\frac{n}{(np+x)[(1-p)n-x]}} \left(\frac{np}{np+x} \right)^{np+x} \left[\frac{n(1-p)}{(1-p)n-x} \right]^{(1-p)n-x}.$$

We take the logarithm of the PDF to find

$$\ln P_n(k) = \ln n! - \ln k! - \ln(n-k)! + k \ln p + (n-k) \ln(1-p).$$

Then, we use Stirling's formula

$$\ln n! = n \ln n - n$$

to find

$$\begin{aligned} \ln P_n(k) &= n \ln n - n - k \ln k + k - (n-k) \ln(n-k) + (n-k) \\ &\quad + k \ln p + (n-k) \ln(1-p) \\ &= -n \ln n - k \ln k + (n-k) \ln(n-k) + k \ln p + (n-k) \ln(1-p) \end{aligned}$$

We seek to use Laplace's method on this. Thus, we differentiate

$$\frac{\partial}{\partial k} \ln P_n(k) = -\ln k + \ln(n-k) - \ln(1-p) + \ln p = 0.$$

Expanding this, we find that

$$k(1-p) = (n-k)p$$

which after rearranging gives

$$k = np$$

as one might expect. Now, we apply Laplace's method, which states that

$$\int_a^b e^{Mf(x)} dx \xrightarrow{M \rightarrow \infty} e^{Mf(x_0)} \int_a^b e^{-\frac{1}{2}M|f''(x_0)|(x-x_0)^2} dx$$

where x_0 is the maximum point of f . We thus expand

$$\ln P_n(k) \approx \ln P_n(np) - \frac{(k-np)^2}{2np(1-p)}.$$

Exponentiating, we find that

$$P_n(k) \propto \exp \left(-\frac{(k-np)^2}{2np(1-p)} \right).$$

This is of the form of a normal distribution. Including the normalisation, we would have

$$P_n(k) = \frac{1}{\sqrt{2\pi np(1-p)}} \exp\left(-\frac{(k-np)^2}{2np(1-p)}\right).$$

In the case $p = \frac{1}{2}$, this is

$$P_n(k) = \sqrt{\frac{2}{\pi n}} \exp\left(-\frac{2(k-\frac{n}{2})^2}{n}\right).$$

- (c) To compute the entropy, we approximate the sum $-\sum P_k \ln P_k$ with an integral $-\int P_k(x) \ln P_k(x) dx$. Since the exponential falls very fast, we can assume that the integration bounds extend to infinity on both sides:

$$S = \int_{-\infty}^{\infty} \sqrt{\frac{2}{\pi n}} \exp\left(-\frac{2(k-\frac{n}{2})^2}{n}\right) \left[\frac{1}{2} \ln \frac{2}{\pi n} - \frac{2(k-\frac{n}{2})^2}{n} \right] dx.$$

□

Problem 2 (Generating Function). A function which can be represented as

$$f(x) = \sum_{k=0}^n a_k x^k$$

is called the generating function of the coefficients a_k .

- (a) Determine the coefficients $\{a_k\}$ for

$$f(x) = (1+x)^n$$

and for

$$f(x) = \frac{1}{1-bx}.$$

- (b) Find the generating function for the coefficients beginning with

$$\{a_k\} = \{0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, \dots\}.$$

There is a nice website at oeis.org. Identify the series and search for “g.f.”, which stands for “generating function”. Verify your result e.g. with Mathematica[®].

- (c) Let $f(x)$, $g(x)$, and $h(x) = f(x)g(x)$ be generating functions with coefficients $\{a_k\}$, $\{b_k\}$, and $\{c_k\}$, respectively. Show that the c_k can be represented as a convolution of the $\{a_k\}$ and $\{b_k\}$. For simplicity ignore the radius of convergence.

Proof. (a) We simply expand them. By the binomial theorem,

$$(1+x)^n = \sum_{k=0}^n \binom{n}{k} x^k.$$

These yield the coefficients directly, i.e.

$$a_k = \begin{cases} \binom{n}{k} & 0 \leq k \leq n \\ 0 & \text{otherwise} \end{cases}.$$

For the function $f(x) = (1-bx)^{-1}$, we apply the binomial series:

$$(1+x)^\alpha = \sum_{n=0}^{\infty} \binom{\alpha}{n} x^n.$$

This directly yields

$$\begin{aligned} f(x) &= (1-bx)^{-1} \\ &= \sum_{n=0}^{\infty} \binom{-1}{n} (-bx)^n \end{aligned}$$

Now, we note that $\binom{-1}{n}$ is just $(-1)^n$, because

$$\binom{-1}{n} = \frac{(-1) \overbrace{(-1-1)}^{-2} \dots \overbrace{(-1-(n+1))}^{-n}}{n!} = (-1)^n.$$

Thus, we have

$$f(x) = \sum_{n=0}^{\infty} (-1)^n (-1)^n b^n x^n = \sum_{n=0}^{\infty} b^n x^n.$$

Thus, the coefficients are given by

$$a_k = b^k.$$

(b) This is the Fibonacci sequence. It is given by the recursion relation

$$a_{k+2} = a_{k+1} + a_k, \quad a_0 = 0, \quad a_1 = 1.$$

Assuming the relevant series converges, we denote the generating function by $G(x)$. We multiply this series with x^k and sum up from $k = 0$ to ∞ :

$$a_{k+2}x^k = a_{k+1}x^k + a_kx^k$$

$$\begin{aligned}
\sum_{k=0}^{\infty} a_{k+2}x^k &= \sum_{k=0}^{\infty} a_{k+1}x^k + \sum_{k=0}^{\infty} a_kx^k \\
\frac{1}{x^2} \sum_{k=0}^{\infty} a_{k+2}x^{k+2} &= \frac{1}{x} \sum_{k=0}^{\infty} a_{k+1}x^{k+1} + \sum_{k=0}^{\infty} a_kx^k \\
\frac{1}{x^2} [G(x) - a_1x - a_0] &= \frac{1}{x} [G(x) - a_0] + G(x)
\end{aligned}$$

Now, we have a simple algebraic equation that we can solve for $G(x)$. We do so easily:

$$\begin{aligned}
G(x) - a_1x - a_0 &= x[G(x) - a_0] + x^2G(x) \\
G(x)[x^2 + x - 1] &= -a_1x + a_0x + a_0 \\
G(x) &= \frac{-a_1x + a_0x + a_0}{x^2 + x - 1} \\
&= \frac{x}{1 - x - x^2}
\end{aligned}$$

(c) We have

$$h(x) = f(x)g(x) = \left(\sum_{n=0}^{\infty} a_nx^n \right) \left(\sum_{m=0}^{\infty} b_mx^m \right) = \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} a_nb_mx^{n+m}.$$

We focus on the coefficient of x^k . To compute this coefficient, we set $m + n = k$. Then, we have $m = k - n$ and

$$c_kx^k = \sum_{n=0}^k a_nb_{k-n}x^k$$

where we have implicitly defined $b_k = 0$ for $k < 0$. Comparing the coefficients, we find that

$$c_k = \sum_{n=0}^k a_nb_{k-n}.$$

This is the desired convolution. □

Problem 3 (Entropy and Lagrange Multipliers). Consider a Markov jump process, where each configuration $c \in \Omega$ is associated with a certain value $\chi_c \in \mathbb{R}$ (for example, the energy).

Use the method of Lagrange multipliers to maximize the entropy

$$H = - \sum_{c \in \Omega} p_c \ln p_c$$

under the constraint that the average value

$$\bar{\chi} = \sum_c p_c \chi_c$$

has a given positive value.

Make sure that your result (the probability distribution p_c) is properly normalized.

Proof. We introduce a Lagrange multiplier λ for the average value and another ζ for the normalisation. We thus have to minimise

$$\mathcal{L} = - \sum_{c \in \Omega} [p_c \ln p_c + \lambda p_c \chi_c + \zeta p_c]$$

with respect to the probabilities p_c , along with the constraint equations

$$\begin{aligned} \bar{\chi} &= \sum_{c \in \Omega} p_c \chi_c \\ 1 &= \sum_{c \in \Omega} p_c \end{aligned}$$

Differentiating with respect to an arbitrary c and demanding that this partial derivative equal to 0, we find the equation

$$0 = \ln p_c + 1 + \lambda \chi_c + \zeta.$$

This leads to the expression

$$p_c = \exp(-1 - \zeta - \lambda \chi_c).$$

Substituting this into the constraints, we find that

$$\begin{aligned} \bar{\chi} &= \exp(-1 - \zeta) \sum_{c \in \Omega} \chi_c \exp(-\lambda \chi_c) \\ 1 &= \exp(-1 - \zeta) \sum_{c \in \Omega} \exp(-\lambda \chi_c) \end{aligned}$$

The second constraint yields

$$\exp(-1 - \zeta) = \frac{1}{\sum_{c \in \Omega} \exp(-\lambda \chi_c)}.$$

Substituting this into the first equation, we find that

$$\bar{\chi} = \frac{\sum_{c \in \Omega} \chi_c \exp(-\lambda \chi_c)}{\sum_{c \in \Omega} \exp(-\lambda \chi_c)}.$$

This implicitly defines the parameter λ ; the probability is then given through

$$p_c = \frac{\exp(-\lambda\chi_c)}{\sum_{c \in \Omega} \exp(-\lambda\chi_c)}.$$

This is the canonical ensemble at inverse temperature λ . □

Problem 4 (Differential Entropy). Let $p(x)$ be a probability density normalized by $\int_{-\infty}^{+\infty} p(x) dx = 1$. Then the corresponding differential entropy is defined by

$$H_c = - \int_{-\infty}^{+\infty} p(x) \ln p(x) dx.$$

Consider a discretization of the x -axis into equidistant intervals of size Δx . Let

$$p_j = \frac{p(j\Delta x)\Delta x}{\mathcal{N}}$$

be a discrete probability distribution approximating $p(x)$, where

$$\mathcal{N} = \sum_{j=-\infty}^{+\infty} p(j\Delta x)\Delta x$$

is the normalization.

- (a) Show that in the limit $\Delta x \rightarrow 0$, the entropy H_c differs from the usual discrete Shannon entropy

$$H = - \sum_j p_j \ln p_j$$

only by an offset

$$\Delta H := H - H_c.$$

- (b) Compute this offset to leading order in Δx .

- (c) Explain your findings qualitatively.

Proof. (a) The discrete Shannon entropy is given by

$$\begin{aligned} H &= - \sum_j p_j \ln p_j \\ &= - \sum_j \frac{p(j\Delta x)\Delta x}{\mathcal{N}} \ln \frac{p(j\Delta x)\Delta x}{\mathcal{N}} \end{aligned}$$

□